

$$\frac{d^2 i(t)}{dt^2} + \frac{R}{L} \frac{di(t)}{dt} + \frac{1}{LC} i(t) = \frac{1}{L} \frac{du_g(t)}{dt}$$

$$\frac{d^2 u_C(t)}{dt^2} + \frac{R}{L} \frac{du_C(t)}{dt} + \frac{1}{LC} u_C(t) = \frac{u_g(t)}{LC}$$

$$\frac{d^2 u_R(t)}{dt^2} + \frac{R}{L} \frac{du_R(t)}{dt} + \frac{1}{LC} u_R(t) = \frac{R}{L} \frac{du_g(t)}{dt}$$

$$\alpha = \frac{R}{2L}$$

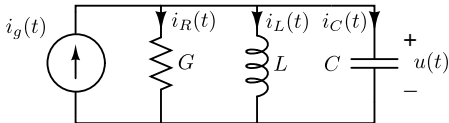
$$\omega_o = \frac{1}{\sqrt{LC}}$$

$$\frac{d^2 f(t)}{dt^2} + 2\alpha \frac{df(t)}{dt} + \omega_o^2 f(t) = g(t)$$

$f(t)$ denota tensión o intensidad.

α , ω_o son constantes del circuito.

$g(t)$ es una función que depende de la excitación.



$$\frac{d^2 u(t)}{dt^2} + \frac{G}{C} \frac{du(t)}{dt} + \frac{1}{LC} u(t) = \frac{1}{C} \frac{di_g(t)}{dt}$$

$$\frac{d^2 i_L(t)}{dt^2} + \frac{G}{C} \frac{di_L(t)}{dt} + \frac{1}{LC} i_L(t) = \frac{i_g(t)}{LC}$$

$$\frac{d^2 i_R(t)}{dt^2} + \frac{G}{C} \frac{di_R(t)}{dt} + \frac{1}{LC} i_R(t) = \frac{G}{C} \frac{di_g(t)}{dt}$$

$$\alpha = \frac{G}{2C}$$

$$\omega_o = \frac{1}{\sqrt{LC}}$$